Human-Computer Interaction (HCI) Quiz Questions

Week 1: Introduction to HCI & Good Design

1. What is the primary goal of Human-Computer Interaction (HCI)?
    - Improve computer hardware
    - **Make interaction effective, efficient, and satisfying**
    - Reduce data usage
    - Teach programming
2. Which example represents bad design?
    - A door with a clear push/pull label
    - **A TV remote with too many buttons**
    - A phone with customizable themes
    - A book with a table of contents
3. Usability mainly describes:
    - App color scheme
    - **How easily users achieve their goals**
    - Internet speed
    - Type of hardware
4. Which of the following is an interactive system?
    - Printed map
    - **Digital ATM touch interface**
    - Wall poster
    - Paper calendar
5. Why is understanding user needs important in user-centered design?
    - To reduce app cost
    - **To match designs with real user behaviors**
    - To create fancy animations
    - To increase marketing budget
6. Why is a door without signifiers (push/pull marks) confusing?
    - It has too much color
    - It uses expensive material
    - **Its affordance and signifiers are unclear**
    - It is automatic
7. What makes a TV remote with many buttons a poor design example?
    - It is heavy
    - Battery drains quickly
    - **Cognitive overload and poor usability**
    - It is not colorful
8. Which step in user-centered design involves understanding user challenges and context?
    - Advertising
    - Deployment
    - **User research**
    - Coding
9. Which HCI principle was violated in the Boeing 737 MAX case?
    - Too many UI colors

    - No touchscreen support
    - **Lack of visibility and poor feedback for automation**
    - Icons were too small
10. Which concept explains why users make errors when controls do not match their mental model?
    - Animation design
    - Aesthetic layout
    - **Mental models and signifiers**
    - Download speed

-----Week 2: User-Centered Design and Interaction Design

1. According to Henry Dreyfuss's idea of design, when does a designer succeed?
    - When the product uses the latest technology
    - **When users feel comfortable and can use the product without friction**
    - When the product is inexpensive
    - When the product looks stylish
2. What was a major reason for the commercial failure of Google Glass?
    - No camera support
    - **Privacy concerns and awkward design**
    - Low screen brightness
    - Lack of wireless connectivity
3. What is the first step in deciding what to design?
    - Finalizing the brand logo
    - **Identifying who the users are**
    - Choosing color schemes
    - Deciding the marketing plan
4. What is the main goal of interaction design according to Sharp, Rogers, and Preece?
    - **Supporting communication and interaction in daily life**
    - Reducing manufacturing costs
    - Increasing screen brightness
    - Designing for entertainment
5. Which of the following best represents User Experience (UX)?
    - **How users feel emotionally while using the product**
    - The processor speed of a device
    - Only the visual layout
    - The number of features in an app
6. Why are multidisciplinary teams valuable in interaction design?
    - They reduce the cost of development
    - They make meetings shorter
    - **They generate diverse ideas and perspectives**
    - They eliminate the need for user testing
7. What is inclusivity in design?
    - Making interfaces look colorful
    - **Designing products for diverse user needs and backgrounds**
    - Removing advanced features
    - Designing only for expert users
8. What is accessibility mainly focused on?

    - Increasing entertainment content
    - Improving device speed
    - **Making products usable by people with disabilities**
    - Reducing UI animations
9. Which design principle ensures that crucial functions are easy to locate?
    - Memorability
    - Flexibility
    - Aesthetics
    - **Visibility**
10. What is affordance in design?
    - The cost of the product
    - **Visual cues that indicate how an element should be used**
    - The brand reputation
    - The color contrast ratio

-----Week 3: Design Process Models and Approaches

1. Which stage of Design Thinking focuses on understanding the user's needs and behaviors?
    - Prototype
    - **Empathize**
    - Test
    - Implement
2. The Double Diamond design process was developed by:
    - Apple
    - Nielsen Norman Group
    - **UK Design Council**
    - Microsoft Research
3. Which of the following is not one of the six stages of Design Thinking?
    - Define
    - Ideate
    - **Evaluate**
    - Empathize
4. In the Define stage, the key goal is to:
    - Sketch wireframes
    - **Frame the problem clearly**
    - Conduct A/B testing
    - Create high-fidelity prototypes
5. Activity-centered design primarily focuses on:
    - User preferences
    - User emotions
    - **Tasks and activities users perform**
    - Visual layout aesthetics
6. Which of the following is not a key component of system design?
    - Data Management
    - Interfaces
    - **Memory recall**
    - Architecture

7. The Interaction Design Process emphasizes four basic activities. Which one of these is included?
    - Monetization
    - **Discovering requirements**
    - Branding
    - Advertising
8. Genius design relies heavily on:
    - Extensive user testing
    - **Designer intuition and expertise**
    - Market surveys
    - Accessibility guidelines
9. Which lifecycle model is based on rapid prototyping and iteration within 5 days?
    - Research in the Wild
    - Waterfall Model
    - **Google Design Sprints**
    - Spiral Model
10. According to Week 3, user involvement is important because:
    - It helps reduce the app size
    - **Users can become stakeholders and contribute feedback**
    - It speeds up coding
    - It avoids the need for iteration

-----Week 4: Understanding Users and Data Gathering

1. User perspectives in HCI help designers understand:
    - Only the visual UI layout
    - **How users perceive and interact with technology**
    - How designers communicate with developers
    - Only system requirements
2. Which of the following is an example of accessibility consideration?
    - Using bright colors only
    - **Providing adjustable font sizes and dyslexia-friendly modes**
    - Adding decorative animations
    - Increasing advertisements
3. Novice users generally prefer:
    - Only voice-based interfaces
    - Highly customizable and complex interfaces
    - Minimal guidance and advanced shortcuts
    - **Simple interfaces with clear instructions**
4. A persona is best described as:
    - A real user hired by the company
    - **A fictional, data-driven representation of a user**
    - A system architecture model
    - A marketing campaign tool
5. Surveys in user research are useful because they:
    - Always provide emotional depth
    - **Allow quick collection of large-scale data**
    - Guarantee unbiased answers

    - Replace interviews completely
6. Empathy mapping categorizes insights into:
    - Features, modules, layouts, workflows
    - **Says, Thinks, Does, Feels**
    - Inputs, outputs, processes, storage
    - Goals, sub-goals, errors, feedback
7. A common challenge in user perspectives is:
    - Too many designers
    - **Conflicting needs among different user groups**
    - Unlimited research time
    - Lack of design tools
8. Affinity mapping is mainly used to:
    - Assign tasks to team members
    - **Organize qualitative data into themes**
    - Remove unnecessary system features
    - Design UI wireframes
9. APOV (Point of View) statement includes:
    - System hardware requirements
    - **User, need, and insight**
    - UI color preferences
    - Coding-level issues
10. Data clustering helps in:
    - **Grouping users based on similar behaviors**
    - Randomly distributing users
    - Increasing device performance
    - Removing prototypes

-----Week 5: Conceptual Models and Cognition

1. Which of the following best describes a conceptual model?
    - A detailed UI layout
    - **A mental representation of how a system works**
    - A list of system requirements
    - A coding guideline document
2. What is an example of an assumption in interface design?
    - Testing a prototype with users
    - **Taking something for granted without evidence**
    - Mapping user journeys
    - Writing code before wireframes
3. Which component is commonly used in online shopping conceptual models?
    - Homepage wallpaper
    - **Shopping cart**
    - Video playback controls
    - Battery indicator
4. Interface metaphors help users by:
    - Making systems slower
    - Increasing hardware complexity
    - **Leveraging familiarity for learning new systems**

    - Removing the need for feedback
5. Which cognitive principle states that decision time increases with more choices?
    - Miller's Law
    - Fitts's Law
    - **Hick's Law**
    - Gestalt Law of Proximity
6. What is the purpose of a system image?
    - To describe internal code structure
    - **To show how the system actually works through the interface**
    - To explain marketing strategy
    - To define business revenue
7. Which of the following is a drawback of interface metaphors?
    - They always reduce cognitive load
    - They break all cultural norms
    - **They may limit designers' creativity**
    - They remove the need for conceptual models
8. Cognitive load increases when:
    - Interfaces use familiar patterns
    - **Users see excessive irrelevant information**
    - Steps are minimized
    - Icons are consistent
9. Which cognitive aspect focuses on how users perceive visual elements?
    - Decision-making
    - **Perception**
    - Problem-solving
    - Reflective cognition
10. Which emerging trend in interfaces directly uses the human brain for input?
    - AR overlays
    - Voice interaction
    - **Brain-Computer Interfaces**
    - Card-based UI

-----Week 6: Data Requirements and Analysis

1. What is the primary purpose of identifying data requirements in HCI?
    - To create marketing campaigns
    - **To determine what information is needed to inform design decisions**
    - To reduce memory usage in applications
    - To choose programming languages
2. Which type of data includes demographics and user preferences?
    - System data
    - **User data**
    - Context data
    - Task data
3. Which of the following is a qualitative data-gathering technique?
    - Analytics logs
    - Surveys with multiple-choice questions
    - **User interviews**

    - System performance metrics
4. Crowdsourcing in HCI is mainly beneficial because:
    - It reduces the need for user testing
    - **It gathers data from diverse and large populations**
    - It guarantees unbiased data automatically
    - It eliminates qualitative research
5. Fairness in data collection ensures that:
    - **All users are treated equitably in data and algorithms**
    - The system runs faster
    - Participants cannot withdraw
    - Only expert users are included
6. Field studies involve:
    - Observing users in controlled lab settings
    - Recording system logs only
    - **Observing users in their natural environment**
    - Removing outliers from datasets
7. Which tool is commonly used for real-time data handling?
    - Tableau
    - **Firebase**
    - NVivo
    - SPSS
8. Which of the following is a key issue in data ethics?
    - Increasing app performance
    - User interface color selection
    - **Transparency in how data is collected and used**
    - Adding more user features
9. Inter-Annotator Agreement (IAA) is used to:
    - Visualize data using graphs
    - **Measure consistency among multiple annotators**
    - Improve system performance
    - Replace manual labeling
10. Which statistical method is used to find relationships between variables?
    - Affinity mapping
    - **Regression analysis**
    - Unstructured interviewing
    - Card sorting

-----Week 7: Prototyping and Smart UI

1. What is a prototype in interaction design?
    - A finalized product
    - **A manifestation of a design that stakeholders can interact with**
    - Only a software demo
    - A marketing document
2. Which of the following is an example of Lo-Fi prototyping?
    - Pixel-perfect UI
    - **Paper sketches**
    - Fully interactive app

    - High-resolution mockup
3. Storyboards in prototyping help designers by:
    - Writing code automatically
    - **Breaking user tasks into scenes to visualize interaction**
    - Generating analytics
    - Creating animations for UI
4. Which prototyping type provides broad coverage with low detail?
    - Vertical
    - **Horizontal**
    - Hi-Fi
    - Wizard-of-Oz
5. Mid-Fidelity prototypes typically include:
    - Stick figures only
    - **Grayscale wireframes with limited interactions**
    - Pixel-perfect color UI
    - Final design animations
6. Smart UI refers to interfaces that:
    - Are built only with code
    - **Use AI to adapt to user behavior and context**
    - Do not require user data
    - Never personalize content
7. Which is an advantage of Smart UI?
    - Random interface behavior
    - **Personalized and adaptive user experience**
    - Increased manual effort for users
    - No support for accessibility
8. Wizard-of-Oz prototyping involves:
    - **A human simulating system responses behind the scenes**
    - Full autonomous system
    - Physical hardware testing
    - Running A/B experiments
9. Vertical prototyping focuses on:
    - A wide range of features with little detail
    - **A detailed exploration of a single feature**
    - Creating only animations
    - Testing final UI only
10. Which tool is commonly used for mid or high-fidelity UI prototypes?
    - Sticky notes
    - **Figma**
    - Lego blocks
    - Scenarios only

-----Week 8: Evaluation and Usability Testing

1. What is the main purpose of evaluation in HCI?
    - To increase hardware performance
    - **To assess usability, effectiveness, and user experience**
    - To test marketing strategies

    - To reduce development cost
2. Which type of evaluation occurs during the design process to refine the product?
    - Summative
    - Diagnostic
    - **Formative**
    - Longitudinal
3. Usability testing primarily involves:
    - Experts reviewing code
    - **Observing real users performing tasks**
    - Only collecting analytics
    - Replacing prototypes with final UI
4. Which evaluation method requires evaluators to simulate user behavior step-by-step?
    - Heuristic Evaluation
    - **Cognitive Walkthrough**
    - A/B Testing
    - Eye Tracking
5. In experiment design, which approach assigns different groups to different conditions?
    - Within-subjects
    - Mixed methods
    - **Between-subjects**
    - Triangulated testing
6. Which metric measures how fast users complete a task?
    - Error severity
    - Task success rate
    - **Time on task**
    - Net Promoter Score
7. Eye tracking helps evaluators understand:
    - System load time
    - **Where users focus attention on the interface**
    - User emotions only
    - The number of app downloads
8. Which principle ensures that participants voluntarily agree to evaluation?
    - Counterbalancing
    - **Informed consent**
    - Aesthetic consistency
    - Reliability
9. Which evaluation method is cost-effective and identifies issues early using established principles?
    - A/B Testing
    - **Heuristic Evaluation**
    - Biometric Metrics
    - Field Studies
10. Which predictive model estimates pointing time based on movement distance and target size?
    - Hick's Law
    - Mental Model

- Cognitive Load Theory
- **Fitts's Law**

-----Week 9: HCI, IoT, and Novel Interaction

1. IoT refers to:
    - **A network of physical objects with sensors and connectivity**
    - Devices that store only offline data
    - Tools without any automation
    - Only entertainment systems
2. The main goal of IoT is:
    - To increase device size
    - **To "connect the unconnected"**
    - To remove automation from homes
    - To limit wireless technology
3. Which example shows HCI + IoT interaction?
    - A robotic arm running without feedback
    - **Users controlling smart bulbs via a mobile app**
    - Devices interacting only through printed manuals
    - A non-digital clock
4. Which interaction modalities are common in IoT?
    - **Voice, gesture, and touch**
    - Only handwritten input
    - Only QR-code scanning
    - Only mechanical switches
5. A major HCI challenge in IoT is:
    - Increasing UI colors
    - **Achieving consistent interaction across many devices**
    - Ensuring devices always remain offline
    - Removing all system notifications
6. Why is HCI crucial in IoT systems?
    - It hides system feedback
    - **It ensures usability and intuitive interaction across devices**
    - It prevents real-time interaction
    - It forces users to follow only one workflow
7. HaptiDrag creates friction using:
    - Heated plates
    - Ultrasonic vibrations
    - **Electroadhesion technology**
    - Physical wheels
8. HaptiDrag enhances VR by:
    - **Giving realistic tactile feedback with variable friction**
    - Removing haptic sensations
    - Reducing VR stability
    - Removing all gestures
9. MobiTangibles supports VR tool training by:
    - **Tracking small tool movements using smartphone magnetometers**
    - Using heat sensors only

    - Removing all sensors
    - Replacing tools with QR codes
10. A limitation of MobiTangibles is:
    - It cannot function on wooden surfaces
    - **It may damage magnetic-sensitive objects like credit cards**
    - It needs no smartphone
    - It cannot detect small magnets at close range

-----Week 10: HCI and AI/LLMs

1. LLMs are primarily designed to:
    - **Predict and generate language using large-scale training data**
    - Render 3D animations
    - Compress multimedia files
    - Control robotic hardware
2. Which of the following is an example of a multimodal input for an AI system?
    - Only typed text
    - **Text, image, or audio combined**
    - A Wi-Fi password
    - A device serial number
3. What is the function of prompting in LLM interactions?
    - Debugging internal model weights
    - **Giving the model instructions or context to perform a task**
    - Measuring CPU temperature
    - Optimizing internet bandwidth
4. Which is a key challenge in AI-based personalization?
    - Limited screen brightness
    - **Privacy concerns around user data**
    - High-fidelity UI design
    - Lack of color contrast in icons
5. In behavior modeling, "Effect" refers to:
    - The color of the UI
    - **The user's actual response or action**
    - The predicted CPU usage
    - The system's refresh rate
6. Zero-shot prompting allows the model to:
    - **Produce outputs without being given examples**
    - Run without electricity
    - Access private user data
    - Train without GPUs
7. Which of these is a benefit of AI for accessibility?
    - Faster firmware updates
    - **Real-time speech-to-text for deaf users**
    - Higher download speeds
    - Larger app icons
8. Behavior-optimized content refers to:
    - **Messages crafted to trigger specific user reactions**
    - Code that compiles automatically

    - Images compressed for mobile phones
    - Deleted browsing history
9. Ethical concerns with LLMs include:
    - Keyboard layouts
    - **Bias, manipulation risk, and misinformation**
    - Slow Wi-Fi speed
    - File naming conventions
10. An MLLM (Multimodal Large Language Model) improves HCI by:
    - **Understanding and generating across multiple input/output formats**
    - Replacing all user research
    - Removing the need for interfaces
    - Turning apps into offline-only tools

ASSIGNMENT 11

1. Privacy in HCI refers to:
   Users' ability to control their personal data
   Increasing app size
   Removing encryption from messages
   Designing only visual elements
2. Which of the following is an example of a security feature?
   Dark themes
   Two-factor authentication (2FA)
   Changing wallpapers
   Offline advertisements
3. Deepfakes are created using:
   Generative Adversarial Networks (GANs)
   Spreadsheet formulas
   Static HTML tags
   Audio compression tools
4. Which type of harm involves AI generating biased or unfair outcomes?
   Bias and discrimination
   Increased RAM usage
   Reduced display brightness
   Slow keyboard response
5. A real-world example of privacy by design is:
   WhatsApp's end-to-end encryption
   Phone ringtones
   Auto-rotate screen settings
   File naming conventions
6. One major risk of deepfakes is:
   Creating propaganda or misinformation
   Improving audio quality
   Faster battery charging
   Removing the need for passwords
7. Security in HCI focuses on:
   Protecting user data from unauthorized access

Increasing the number of UI animations
Designing only for children
Removing all login screens

8. GDPR impacts HCI design by requiring:
   Clear consent mechanisms and privacy transparency
   More emojis in interfaces
   Mandatory dark mode
   Faster loading animations
9. Which tool supports secure authentication?
   Google Authenticator
   Stopwatch app
   Video filters
   Calculator app
10. A major challenge in privacy & security design is:
    Balancing usability with strong protections
    Removing all security features entirely
    Hiding all system messages
    Making interfaces more confusing

ASSIGNMENT 12

1. Shannon's Level C focuses on:
   Whether the received meaning achieves the intended effect
   Hardware choices
   Signal compression
   Gesture selection
2. In human communication, encoding means:
   Choosing fonts
   Converting thoughts into words or symbols
   Measuring noise levels
   Compressing audio
3. A key feature of modern conversational AI is:
   Natural Language Understanding
   Monochrome output
   Manual rule-writing
   Offline-only processing
4. An example of a conversational AI assistant is:
   A calculator
   A stapler
   Google Assistant
   A desk lamp
5. A communication barrier includes:
   Wallpapers
   Volume changes
   Extra brightness
   Semantic misunderstandings

6. AI conversational systems typically:
   Only record audio
   Use NLP to interpret user messages
   Require physical switches
   Ignore user queries
7. A challenge in deploying conversational AI is:
   Bias and misunderstanding nuanced input
   Limited font choices
   Keyboard shortcuts
   Battery drainage
8. Asimov's First Law states:
   Robots must obey all commands
   Robots must protect themselves
   Robots may not harm humans or allow harm
   Robots must avoid emotions
9. Socially responsible MIAs aim to:
   Compress images
   Promote fairness and inclusive interactions
   Reduce accessibility
   Avoid cultural sensitivity
10. A benefit of conversational AI in healthcare is:
    Changing screen colors
    Increasing room volume
    Printing receipts
    Offering virtual guidance such as symptom checking